MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 3 of 10 (Start time: 03:11:00 PM, 17 June 2025)

Is the function f(x) = tan (x) continuous at x = pi ? If not, why?

Select the correct option

- (●) f is continuous at x = pi .
- ( ) f (pi) is not defined.

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Question # 4 of 10 (Start time: 03:11:20 PM, 17 June 2025)

The sum of two differentiable functions is always differentiable.

Select the correct option

○	False
◉	True

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 6 of 10 (Start time: 03:15:59 PM, 17 June 2025)

The derivative of Sin(3x) at the x=0 is _____.

Select the correct option

○	-3
●	3
○	2
○	-1

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 1 of 10 (Start time: 03:14:06 PM, 17 June 2025)

What is the value of $\lim_{x \to 0} \frac{sinx^2}{x^2}$?

Select the correct option

○	0
○	-1
○	∞
◉	1

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Question # 7 of 10 (Start time: 03:12:17 PM, 17 June 2025)

$\lim_{h \to 0} f(x) = ?$

Select the correct option

○	$f(x-h)$
○	$f(x+h)$
○	0
◉	$f(x)$

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Question # 7 of 10 (Start time: 03:16:12 PM, 17 June 2025)

If the function is differentiable at a point then it is also _____ at that point.

Select the correct option

○	Not defined
◉	Continuous

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Question # 5 of 10 (Start time: 03:11:41 PM, 17 June 2025)

Let y = x^2+1
then the average rate of y with respect to x over the interval [2,1] is

Select the correct option

◉	-3
○	3
○	5
○	-5

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 8 of 10 (Start time: 03:16:27 PM, 17 June 2025)

Let f(x)=Cos(x)/6 , the derivative of f(x) is

Select the correct option

○	-sinx/0
◉	-sinx/6
○	sinx
○	6sinx

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 1 of 10 (Start time: 03:10:24 PM, 17 June 2025)

Tan(x) is continuous everywhere except at

Select the correct option

○	$sin(x) = 0$
●	$cos(x) = 0$
○	$cos(x) \neq 0$
○	$sin(x) \neq 0$

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Question # 3 of 10 (Start time: 03:15:07 PM, 17 June 2025)

A bus travels 175 miles in 2.5 hours. Its average velocity will be.....

Select the correct option

○	60m/h
○	80m/h
○	90m/h
◉	70m/h

Question # 9 of 10 (Start time: 03:16:41 PM, 17 June 2025)

If $f(x) = x^5 + 10$, then which of the following is true about it?

Select the correct option

○	None of these.
○	Its derivative is 10.
◉	Its derivative is $5x^4$.
○	Its derivative is x^4.

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Question # 5 of 10 (Start time: 03:15:43 PM, 17 June 2025)

If the slope of a function is zero, then which of the following is true about it.

Select the correct option

○	None of these.
●	The function should be a constant.
○	The function should be a quadratic.
○	The function should be a linear.

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 2 of 10 (Start time: 03:10:44 PM, 17 June 2025)

If f, g and h are functions satisfying $f(x) \leqslant g(x) \leqslant h(x)$. Also $\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = 3$, then $\lim_{x \to a} g(x) =$

Select the correct option

○	$f(x)$
○	a
●	3
○	$h(x)$

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 8 of 10 (Start time: 03:12:42 PM, 17 June 2025)

If a car travels 150 miles over a straight road in 3 hours, then its average velocity during the trip is .....

Select the correct option

○	60m/h
●	50m/h
○	30m/h
○	40m/h

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 6 of 10 (Start time: 03:11:54 PM, 17 June 2025)

Consider the function defined by $\frac{1}{x^2-x-6}$, which is continuous everywhere_____.

Select the correct option

○	$except\ x=2, x=3$
●	$except\ x=-2, x=3$
○	$except\ x=-2, x=-3$
○	$except\ x=2, x=-3$

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 10 of 10 (Start time: 03:16:57 PM, 17 June 2025)

What is the derivative of $2x^2 \sin(x^5)$?

Select the correct option

◉	None of these
○	$10x^2 \cos(x^5)$
○	$-10x^2 \cos(x^5)$
○	$2x^2 \cos(x^5)$

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 4 of 10 (Start time: 03:15:20 PM, 17 June 2025)

What is the derivative of $2\tan(3x)$?

Select the correct option

○	$-6sec^2(3x)$
○	None of these
◉	$6sec^2(3x)$
○	$3sec^2(3x)$

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 9 of 10 (Start time: 03:12:55 PM, 17 June 2025)

If f(x) approaches to L as x approaches to a from left and right, then f(x) is in the interval

Select the correct option

◉	(L- e , L+ e) where e is a small positive number.
○	Not defined
○	(L- 1 , L+ 1)
○	(-8 , +8)

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 10 of 10 (Start time: 03:13:13 PM, 17 June 2025)

What does the statement $\lim_{x \to a} f(x) = L$ *mean?*

Select the correct option

○	The maximum value of $f(x)$ occurs at $x=a$.
◉	As x approaches a, the values of $f(x)$ get arbitrarily close to L.
○	The value of $f(x)$ at $x=a$ is equal to L.
○	The average rate of change of $f(x)$ is L near x=a.

MTH101 - Calculus And Analytical Geometry (Quiz No. 2)

Question # 2 of 10 (Start time: 03:14:32 PM, 17 June 2025)

$\frac{d}{dx}(f(x)g(x)) = ?$

Select the correct option

○	$\lim_{h\to 0}\frac{f(x+h)-g(x)}{h}$
●	$\lim_{h\to 0}\frac{f(x+h)g(x+h)-f(x)g(x)}{h}$
○	$\lim_{h\to 0}\frac{g(x+h)-f(x)}{h}$
○	$\lim_{h\to 0}\frac{f(x)g(x+h)-f(x+h)g(x)}{h}$